

we refer to as the **Alignment Instability Condition**, that explains why these critical parameters are susceptible to drift even when the ecosystem of open-model development is expanding rapidly [40].

Theoretical Frameworks for Skill Emergence and Adaptation. Our decomposition of alignment into distinct utility functions is grounded in recent theoretical advances regarding skill emergence in large language models. [4] formally model this phenomenon, proposing that complex skills emerge from the scaling of random feature combinations that allow the model to distinguish between increasingly granular tasks. This skill-based view is further supported by [37], who provide a kernel-based analysis of how fine-tuning modifies these underlying feature mappings. Our work adopts this foundational perspective by viewing alignment features not as global constraints but as distinct, emergent skills. However, we reframe the problem from acquisition to preservation. While [4] explain how these high-dimensional skills arise, our geometric theory addresses the inverse problem: why specific skills possess a unique structural fragility that makes them susceptible to catastrophic forgetting even when general capabilities remain intact. This fragility is further understood by examining the intersection of our work with meta-learning theory. The standard paradigm for multi-task learning, as established in [19, 56], posits that diverse tasks can be solved by learning a single, shared low-dimensional representation, upon which task-specific linear heads are trained. In this idealized setting, fine-tuning is safe because the underlying representation is fixed. However, practical fine-tuning violates this assumption by updating the representation itself. [11] address this by relaxing the requirement to an approximate shared representation, where task-specific parameters are constrained to remain within a small distance of the central meta-parameters. While [11] prove that this relaxation allows for effective adaptation, our AIC reveals a critical blind spot in this framework regarding safety. We demonstrate that on curved manifolds, the allowable deviation cannot be isotropic. Due to the high curvature associated with alignment skills, even valid updates that satisfy the distance constraint for a new task can induce large, non-linear drifts in the safety subspace. Thus, our work challenges the sufficiency of the approximate shared representation hypothesis for safety-critical systems, showing that geometric curvature couples benign task updates with more harmful alignment degradation in ways that standard meta-learning bounds do not predict.

3 Model & Preliminaries

We consider a language model parametrized by $\theta \in \Theta \subseteq \mathbb{R}^n$, defining a conditional distribution, or policy, $\pi_\theta(y|x)$ over responses $y \in \mathcal{Y}$ given prompts $x \in \mathcal{X}$. Unless otherwise stated, we let $\|\cdot\|$ denote the Euclidean norm of a vector and $\|\cdot\|_{\text{op}}$ denote the operator norm.

3.1 Alignment as Skill Utility

A key obstacle to theoretical analysis is that alignment is not monolithic. A model simultaneously maintains multiple capabilities that can degrade independently during fine-tuning. We therefore decompose alignment into distinct *skills* (or capabilities), following recent work on skill emergence in language models [4, 37].

Definition 3.1 (Alignment Skill and Utility). *Let $\mathcal{S} = \{S_1, \dots, S_m\}$ be a set of alignment skills. Each skill S_i is associated with a reference distribution $\mathcal{D}_i$ over interaction pairs (x, y) . We define the **utility** of the model θ for skill S_i ,*

$$u_i(\theta) := \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\log \pi_\theta(y|x)] .$$

For a fixed base model, θ^* , we define the utility loss (or alignment degradation) for skill S_i under parameter update $\Delta\theta$ as $\Delta u_i(\theta) := u_i(\theta^*) - u_i(\theta)$. This definition grounds model alignment in probabilistic terms: a model is well-aligned for skill S_i if it assigns high likelihood to responses deemed ideal under $\mathcal{D}_i$.

3.2 Local Geometry and Fisher Information

To analyze alignment preservation geometrically, we adopt an idealized but analytically convenient assumption that the base model is well-optimized for the alignment skills under consideration.

Assumption 1 (Skill Optimality). *The base model θ^* is optimal for every alignment skill $S_i: \pi_{\theta^*}(y|x) = \mathcal{D}_i(y|x)$ for all $x \in \text{supp}(\mathcal{D}_i)$ and all y with $\mathcal{D}_i(y|x) > 0$.*

This assumption states that prior to fine-tuning, the base model exactly satisfies the developer-specified alignment objectives. While idealized, similar well-defined assumptions are standard in theoretical analyses to obtain clean local characterizations of learning dynamics. In our setting, Assumption 1 ensures that θ^* is stationary for each skill’s utility and that local alignment degradation admits a transparent reduction to a KL divergence [14]. We emphasize that Assumption 1 is stronger than necessary and is adopted for clarity of presentation. Our main results extend under weaker local conditions (see Appendix C).

Lemma 3.2 (Alignment Loss as KL Divergence). *Given Assumption 1, after fine-tuning to obtain model θ , the degradation in utility for any skill S_i is equal to the expected KL divergence between the policies, measured over the distribution $\mathcal{D}_i$:*

$$\Delta u_i = \mathbb{E}_{x \sim \mathcal{D}_i(x)} [\text{D}_{\text{KL}}(\pi_{\theta^*}(y|x) || \pi_{\theta}(y|x))].$$

The proof follows from the definition of utility and is deferred to Appendix A. This result highlights that alignment stability is determined by the model’s policy divergence on the *alignment* distribution $\mathcal{D}_i$, which may differ significantly from the fine-tuning distribution. In practice, alignment comprises multiple skills (safety refusal, medical accuracy, factual correctness, etc.). Our analysis focuses on individual skill degradation, where overall alignment fails when any critical skill degrades below acceptable thresholds.

Trust Regions and Local Quadratic Geometry. A common safeguard, such as those in PPO [49] or TRPO [48], constrain updates within a KL trust region around the base policy:

$$D_{\text{KL}}(\pi_{\theta^*} || \pi_{\theta}) \leq \Delta,$$

where $\Delta > 0$ is a small positive constant. While such constraints were initially designed for optimization stability, in alignment settings they are often interpreted as preventing harmful deviation from a safe reference model [62]. Geometrically, this restricts fine-tuning to a neighborhood where the parameter space is locally quadratic. We quantify this curvature using the Fisher Information Matrix (FIM). Let $\mathbf{F}_x(\theta^*) = \mathbb{E}_{y \sim \pi_{\theta^*}(\cdot|x)} [\nabla_{\theta} \log \pi_{\theta^*}(y|x) \nabla_{\theta} \log \pi_{\theta^*}(y|x)^{\top}]$ be the Fisher matrix for input x . We further define the FIM specific to skill S_i as $\mathbf{F}_i(\theta) := \mathbb{E}_{x \sim \mathcal{D}_i} [\mathbf{F}_x(\theta)]$. Assuming the model is C^3 smooth, the second-order Taylor expansion of the KL divergence yields the following local form for alignment loss:

Proposition 3.3 (Local Geometric Form). *For sufficiently small displacements $\Delta\theta = \theta - \theta^*$,*

$$\Delta u_i = \frac{1}{2} \Delta\theta^{\top} \mathbf{F}_i(\theta^*) \Delta\theta + O(\|\Delta\theta^3\|). \quad (1)$$

Thus, in the small-step regime enforced by KL trust regions, skill degradation is locally dominated by the projection of the update $\Delta\theta$ onto the high-curvature directions of $\mathbf{F}_i(\theta^*)$.

3.3 Low-Rank Sensitivity Structure

The key geometric insight is that not all directions in parameter space affect alignment equally. Empirical studies of neural network loss landscapes reveal that Hessians (and FIMs) typically have spiked spectra: a few large eigenvalues dominate, while most directions have near-zero curvature [47]. Recent empirical work also shows that model weights share a low-rank joint subspace across different tasks [32]. This further motivates a low-rank decomposition of alignment sensitivity.

Definition 3.4 (Alignment Sensitivity Subspaces). *The alignment sensitivity subspace M_i for skill S_i is the span of the d -leading eigenvectors of $\mathbf{F}_i(\theta^*)$.*

Stemming from this definition, let $P_i : \mathbb{R}^n \rightarrow M_i$ denote the Euclidean orthogonal projection onto M_i , and $P_i^{\perp} = I - P_i$, the projection onto its orthogonal complement. Within our framework, directions in M_i are critical for alignment, with small movements incurring large utility loss due to high curvature, whereas directions in the null space are locally benign. This decomposition formalizes the empirical observations that alignment resides in small subspaces [5, 35, 60].

4 Why Flat Geometry Fails

Section 3 established a local geometric picture: utility loss is controlled by projections onto low-dimensional sensitivity subspaces, and KL trust regions restrict fine-tuning to neighborhoods where this quadratic approximation holds. This suggests a powerful alignment-preservation guarantee: in high dimensions, unrelated fine-tuning updates should be nearly orthogonal to any fixed low-rank subspace, making degradation exponentially unlikely. We formalize this intuition as a probabilistic **null model**, then contrast it against empirical evidence showing that real fine-tuning trajectories violate these predictions systematically.

4.1 Benign Fine-Tuning in the Flat Regime

To model benign fine-tuning without additional structure, we idealize the update direction as a random unit vector in parameter space.

Theorem 4.1 (Benign Fine-Tuning under Flat Geometry). *Fix an alignment skill S_i and assume there exists $r > 0$ such that for all $\|\Delta\theta\| \leq r$, we have that $\Delta u_i = \frac{1}{2}\Delta\theta^\top \mathbf{F}_i(\theta^*)\Delta\theta + O(\|\Delta\theta\|^3)$. Let $g \sim \text{Unif}(\mathbb{S}^{n-1})$ be a random unit vector, and consider the update $\Delta\theta = \eta g$ where $\eta > 0$ is sufficiently small. Then: (1) the expected utility loss satisfies*

$$\mathbb{E}_{g \sim \text{Unif}(\mathbb{S}^{n-1})} [\Delta u_i] = \frac{\eta^2}{2n} \text{Tr}(\mathbf{F}_i(\theta^*)) + O(\eta^3),$$

(2) for any rank- d orthogonal projector P , we have $\|P(g)\|^2 = \Theta(d/n)$, with probability at least $1 - 2e^{-c\epsilon^2 d}$. This further implies that

$$\Delta u_i = O\left(\frac{d\eta^2}{n}(1 + \epsilon)\lambda_{\max}(\mathbf{F}_i(\theta^*))\right).$$

The proof uses rotation invariance of the uniform distribution on the sphere and concentration of χ^2 random variables (Appendix A).

This theorem formalizes the high-dimensional safety intuition: when the utility loss is well-approximated by a fixed quadratic form and fine-tuning updates are unstructured, the expected degradation scales with the average curvature $\text{Tr}(\mathbf{F}_i(\theta^*))/n$ and the probability of significant overlap with any fixed low-dimensional sensitivity subspace is exponentially small in its dimension. In this regime, alignment degradation is suppressed both in expectation and with high probability by the ambient dimensionality of the parameter space. Consequently, if harmful or alignment-critical directions are confined to a small subspace and fine-tuning updates are effectively random, narrow fine-tuning should be benign. However, this conclusion rests on two idealizations: (1) updates are unstructured (modeled as random), and (2) the Fisher geometry is static throughout training. As we show next, both assumptions fail in practice.

4.2 Empirical Alignment Drift

A growing body of evidence contradicts the null model’s predictions. Alignment degradation is not rare, and is, in fact, common, rapid and strongly dependent on task structure in ways not explained by random projection.

Benign Tasks Break Safety. The strongest challenge to the predictions of Theorem 4.1 comes from fine-tuning on datasets with no harmful content. [25] demonstrate that fine-tuning LLaMa-2-7B and GPT-3.5-Turbo on standard instruction datasets for fewer than 100 steps causes sharp drop off (within five gradient steps) in safety refusal on adversarial prompts, consistent with the results of [20]. Their work further shows that even carefully filtered data can trigger such jailbreaks with only 100 benign instruction-following samples. [6] and [44] further discuss this “emergent misalignment” by showing that narrow task adaptation systematically erodes safety boundaries.

Non-Uniform Sensitivity and Threshold Effects. If degradation arose from random overlap as Theorem 4.1 predicts, we would expect smooth, gradual decline proportional to training time. Instead, [60] observe sharp transitions, where modifying a specific 3% of weights causes catastrophic safety collapse, while random 3% samples have negligible effect. This non-uniformity directly contradicts the null model’s assumption that degradation scales with average curvature across all parameters. Moreover, [44] and [57] document threshold behaviors where models remain stable for several steps before rapid failure onset, again inconsistent with smooth probabilistic degradation.